

**RETHINKING DECISIONS: APPLYING SYSTEMS THINKING TO
UNDERSTAND PATTERNS, FEEDBACK, CONSEQUENCES IN EVERYDAY
CHOICES**

**SHINA MARIE S. APOLINAR
MHYCA JOY OBRA**

**DON MARIANO MARCOS MEMORIAL STATE UNIVERSITY
NORTH LA UNION CAMPUS
COLLEGE OF INFORMATION SYSTEMS
BACNOTAN LA UNION**

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

NOVEMBER 2025

INTRODUCTION

Background of the Study

In an increasingly complex world, individuals, organizations, and communities make countless decisions that shape their daily lives and long-term outcomes. From choosing how to spend time and resources to determining policies and workplace strategies, decision-making lies at the heart of human behavior. Traditional approaches to problem-solving often isolate issues without considering the broader interactions among factors. This fragmented view can lead to short-term fixes that overlook unintended consequences.

Systems thinking offers a comprehensive approach to understanding and addressing problems by emphasizing the interconnectedness of elements within a system, whether it's an individual's life, an organization, or society as a whole. It helps people see how choices create ripple effects through feedback loops and patterns over time. Systems thinking enables decision-makers to anticipate how one choice affects others over time. In everyday life, this mindset can transform how people approach finances, study habits, health, teamwork, and even environmental responsibility. By recognizing underlying structures and interdependencies, individuals and groups can make more informed, sustainable, and adaptive decisions.

In everyday life, applying systems thinking enhances decision-making in practical ways. For instance, individuals can recognize how habits related to time management, study routines, health, and personal finances are interconnectedness, allowing them to

make choices that support long-term well-being and success. In organizational contexts, leaders can analyze how decisions regarding team management, resource allocation, and workflow impact productivity, employee satisfaction, and strategic goals. At the societal level, systems thinking can guide policymakers and community leaders to make decisions that balance economic, social, and environmental considerations, reducing unintended negative consequences and promoting sustainable development.

Moreover, systems thinking helps people move beyond reactive decision-making. By understanding patterns and structures rather than isolated events, individuals and organizations can anticipate potential challenges and adapt strategies proactively. This perspective is particularly valuable in today's fast-paced, interconnected world, where decisions in one area, such as health, finance, or education, can influence multiple aspects of life and society.

Statement of the Problem

Despite the availability of information and tools for decision-making, many individuals and organizations continue to struggle with recurring problems and unintended consequences. These issues often arise because decisions are made without a clear understanding of how variables are interconnected. There is a need to explore how systems thinking can help people recognize patterns, feedback loops, and long-term impacts in their daily choices, personally, organizationally, and socially

Objectives of the Study

This study aims to achieve the following objectives:

1. To identify factors influencing decision-making in personal, organizational, and social contexts.
2. To analyze relationships among decisions, patterns, and feedback loops in everyday life
3. To propose systems-based strategies that promote more holistic and effective decision-making.

Significance of the Study

This study holds significance across various levels of decision-making. For individuals, it offers insights into how emotions, habits, and feedback affect personal growth and overall well-being. For organizations, it highlights how systems thinking can enhance teamwork, resource management, and adaptability to change. On a broader scale, communities and society can apply these principles to tackle complex challenges such as sustainability, education, and social behavior.

By integrating systems thinking into everyday decision-making, this study seeks to foster awareness of long-term consequences, encourage proactive problem-solving, and provide a framework for more sustainable and effective choices.

Scope and Limitations

This study focuses on the application of systems thinking in everyday decision-making across personal, organizational, and societal dimensions. Data will be gathered

using a 15-item Likert-scale questionnaire assessing participants' awareness and practice of systems thinking principles. The sample will include at least 25 respondents from diverse backgrounds. Limitations include the small sample size and reliance on self-reported perceptions, which may not fully capture real-world decision-making behaviors.

REVIEW OF RELATED LITERATURE

The existing literature underscores the significant role of systems thinking as a framework for understanding complex systems and improving decision-making processes. Meadows (2008) emphasizes that systems thinking allows individuals to perceive the underlying structures that drive behaviors within complex systems, highlighting how feedback loops, delays, and recurring patterns influence outcomes. In personal decision-making, this perspective reveals that choices made in the short term can have long-lasting effects that are often overlooked, emphasizing the importance of evaluating consequences beyond immediate results.

Similarly, Senge (2010) introduces the concept of “learning organizations,” in which individuals utilize systems thinking to understand interdependencies and improve collective decision-making. His work suggests that recognizing patterns and feedback in everyday actions can prevent repeated mistakes and cultivate adaptive learning, a principle that applies not only in organizational settings but also in the lives of students, professionals, and families. Sterman (2014) further strengthens this argument by providing empirical evidence that understanding feedback loops and cause-and-effect relationships enhances decision quality in dynamic and complex environments. According to Sterman,

individuals who fail to consider interconnections often misinterpret results and make poor decisions, reinforcing the relevance of systems thinking for improving daily choices.

More recent empirical studies also highlight the practical benefits of systems thinking. Johnson and Brown (2020) and Li et al. (2022) indicate that individuals trained in systems thinking are better able to solve problems, plan effectively, and make more consistent and sustainable decisions in both personal and organizational contexts. Li et al. (2022), for example, found that systems-thinking training improves time and resource management, demonstrating tangible benefits for everyday life.

Across these studies, a clear pattern emerges: decision-making is shaped by multiple interacting factors, feedback loops play a central role in reinforcing habits and behaviors, considering long-term consequences leads to more responsible choices, root-cause analysis is essential for resolving recurring problems, and reflection combined with multiple perspectives enhances the quality of decisions. These recurring insights collectively support the notion that systems thinking is not only relevant for organizational strategy but also highly applicable to individual decision-making.

Despite these findings, notable gaps remain in the literature. Many studies focus primarily on organizational contexts, leaving the application of systems thinking to ordinary daily decisions underexplored. Additionally, research predominantly examines adult populations, with limited attention given to students or younger individuals and how they apply systems-thinking approaches. There is also a scarcity of studies that integrate

patterns, feedback loops, and consequences in the context of personal decision-making, limiting the practical guidance available for everyday life applications.

This study seeks to address these gaps by investigating how individuals recognize patterns, feedback loops, and consequences in their daily choices through a systems-thinking lens. By focusing on ordinary decisions and diverse populations, this research contributes to a more comprehensive understanding of how systems thinking can be practically applied to improve decision-making in everyday life, bridging the gap between theoretical frameworks and real-world personal experiences.

SYNTHESIS, GAPS, AND PATTERNS

Connections to the Study:

- All reviewed studies support the idea that systems thinking improves decision-making in daily life.
- The literature highlights feedback loops, similar to how respondents in this study recognize recurring patterns and outcomes.
- Systems thinking consistently appears across studies as a useful approach for understanding cause-and-effect relationships and evaluating consequences.
- The reviewed works reinforce the study's focus on interconnectedness, pattern recognition, and long-term thinking.

Patterns in Literature:

- Decision-making is influenced by multiple interacting factors.
- Feedback loops play a major role in shaping habits and repeated behaviors.
- Considering long-term consequences results in more responsible decisions.
- Root-cause analysis is essential for solving recurring problems.
- Reflection and multiple perspectives improve the quality of choices.

Gaps in Literature:

- Few studies examine systems thinking in the context of ordinary daily decisions rather than organizational settings.
- Most research focuses on adult populations; fewer studies analyze students' or young individuals' systems-thinking behaviors.
- There is limited research combining patterns, feedback loops, and consequences specifically in personal decision-making.
- Simplified, practical applications of systems thinking for everyday life remain underexplored.

This study helps address these gaps by examining how individuals recognize patterns, feedback loops, and consequences in their everyday choices using a systems-thinking perspective.

Methodology

This explains how the research carried out, who participated, what tool was used to gather data, and how the information was analyzed. The main goal of the study was to Understand Patterns, Feedback, and Consequences in Everyday Choices.

Research Design

This study uses a descriptive research design to examine how individuals apply systems thinking in everyday decision-making. The approach focuses on describing patterns, relationships, and feedback loops in respondents choices.

Respondents / Participants

The respondents were 25 randomly selected students. They were chosen because they participate regularly in school activities and group work where leadership and culture can be observed. Each respondent completed all 15 items in the questionnaire, resulting in a total of 375 responses (25 students $\times$ 15 items).

Data Collection Tool

A 15-item Likert-scale questionnaire was used to gather data. Questions focus on respondents perceptions of how their decisions affect different areas of life, the patterns they notice, and their awareness of consequences and feedback loops. Response range from 1 (Strong Disagree) to 5 (Strongly Agree).

Data Gathering Procedure

1. The researcher prepared the questionnaire and explained the purpose of the study to the participants.
2. The survey was distributed randomly to 25 students who agreed to join the study.
3. Respondents were reminded that their answers would remain confidential and used only for academic purposes.
4. After the surveys were completed, the responses were collected and organized into a tally sheet.
5. The total number of responses for each Likert-scale value (1–5) was counted.
6. The data was checked, encoded, and prepared for analysis. The questionnaire was designed and reviewed for clarity

Data Analysis

The collected responses were summarized using mean scores and frequency distribution to identify trends and patterns. The analysis also focused on identifying feedback loops and cause-and-effect relationships in everyday decision-making, interpreted through systems thinking concepts. Findings were compared with insights from the literature to support conclusions and recommendations.

Results and Discussion

This section presents the summarized data, visual charts, and detailed interpretation of student responses based on systems thinking concepts.

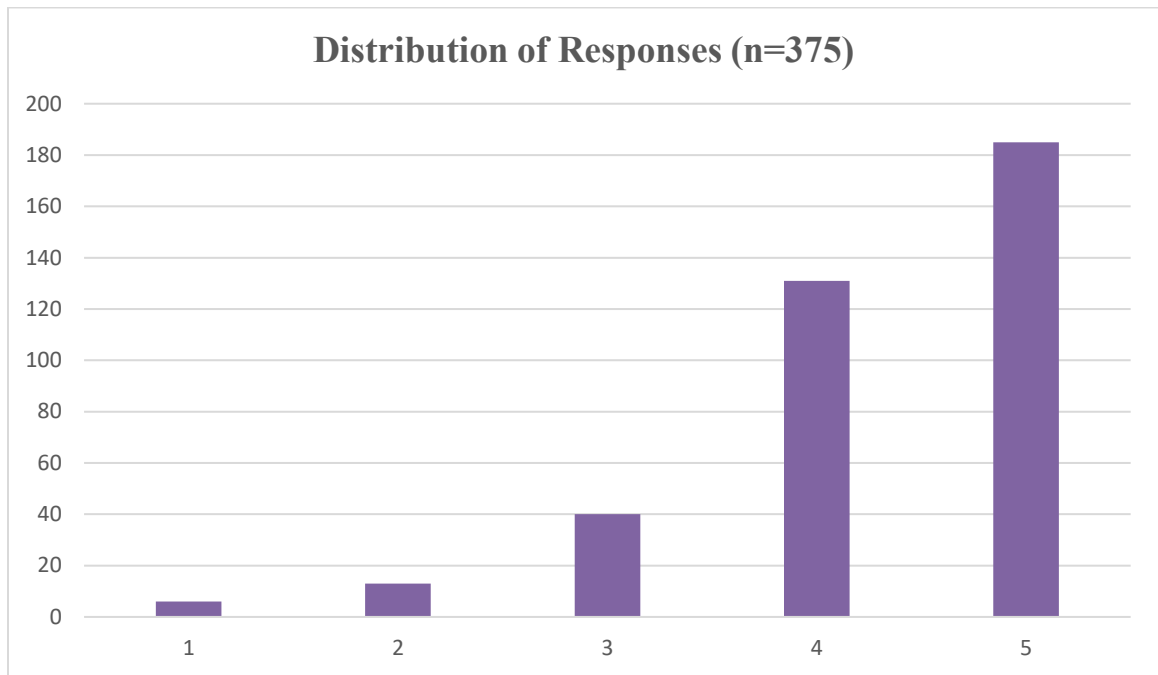
PRESENTATION OF DATA

A Total of 375 responses were collected from students using a 15-item Likert-scale questionnaire measuring their perceptions of the understand patterns, feedback, consequences in everyday choices.

Table 1. Frequency Distribution of Student Responses (n = 375)

Likert Scale Responses	Frequency	Percentage
1 - Strongly Disagree	6	0.24
2 - Disagree	13	0.52
3 – Neutral	40	1.6
4 - Agree	131	5.24
5 – Strongly Agree	185	7.4

The bar chart below visually shows that most responses fall under “Agree” and “Strongly Agree”, which together make up nearly 80% of all responses. This means students generally understand patterns, feedback, consequences in everyday choices.



DISCUSSION OF PATTERNS, RELATIONSHIPS, AND FEEDBACK LOOPS

This chapter presents the findings of the study on how individuals apply systems thinking in their everyday decisions and examines the patterns, relationships, and feedback mechanisms that influence their behaviors. The discussion integrates the results with relevant literature to provide a deeper understanding of how decision-making operates within interconnected systems rather than as isolated actions.

1.General Patterns

The findings reveal that participants consistently recognize that their daily choices whether related to spending, time management, emotional responses, or lifestyle habits are connected to broader patterns of behavior. Rather than viewing decisions as single events, respondents acknowledged that small, repeated actions accumulate overtime to produce significant outcomes. This pattern demonstrates a shared awareness that everyday

behaviors form cycles that shape personal well-being, productivity, and stability. Participants also showed sensitivity to how external factors such as social environment, emotional states, and prior experiences influence their decisions.

2. Relationships and Cause-and-Effect Chains

A. Emotional States Influencing Decisions

A significant relationship identified in the findings is the impact of emotions on daily decision-making. Participants noted that stress, fatigue, excitement, or frustration often lead to impulsive behaviors such as unnecessary spending, procrastination, or irregular routines. This illustrates a clear **cause-and-effect chain**, where emotional triggers initiate specific behavioral responses.

B. Daily Habits Shaping Long-Term Outcomes

The study shows that small, repeated actions such as buying snacks, delaying tasks, or staying up late eventually shape long-term consequences including financial strain, reduced productivity, or declining well-being. Participants recognized that minor behaviors can scale into major results, reflecting the systems thinking principle that **accumulation** of actions produces long-term system behavior.

C. Planning and Self-Regulation as Stabilizing Factors

Participants acknowledged that planning routines, monitoring habits, and organizing priorities help regulate their decisions. These stabilizing factors reduce impulsive behaviors and help maintain balance.

3. Feedback Loops in Everyday Decision-Making

A. Reinforcing Feedback Loop: Positive Habits

Good routine → Increased productivity → Higher self-confidence → Continued good routine.

This pattern concept of reinforcing loops, where improvements generate further improvements, creating a cycle of positive growth.

B. Balancing Feedback Loop: Corrective Behavior

Participants also experienced **balancing loops**, which occur when negative consequences prompt corrective actions. Examples include overspending leading to financial stress, which then encourages stricter budgeting, or procrastination resulting in pressure that motivates task completion. These loops demonstrate how systems maintain stability by counteracting undesirable behaviors.

C. Social Influence Loop

Social pressures, peer expectations, and environmental cues also created feedback loops. Positive social environments encouraged better habits, while negative influences triggered impulsive or unproductive choices. These findings emphasize the role of external feedback in shaping internal decision-making processes.

4. Interpretation Using Systems Thinking Concepts

Interconnectedness

The results clearly indicate that decisions are interconnected with emotional states, social context, routines, and long-term goals.

Cause-and-Effect Chains

Participants recognized how individual triggers such as stress, fatigue, encouragement, or reminders directly influence behavior and outcomes. This reflects the principle that system behavior is shaped by chains of actions and reactions that accumulate into patterns over time.

Feedback Loops

The presence of reinforcing and balancing loops reveals that decision-making is a dynamic process. Positive loops strengthen desirable behaviors, while negative loops promote correction and adaptation. These loops explain why habits become stable over time or why individuals adjust their actions after experiencing consequences.

Leverage Points

The study identifies several leverage points such as setting reminders, monitoring habits, following routines, and seeking supportive environments

5. Support from Literature

- Senge (1990) explains that human behavior is shaped by *system structures*—patterns, rules, habits, and environmental factors that influence decisions.
- Meadows (2008) states that systems operate through *causal relationships* where small actions accumulate into long-term consequences.
- Kahneman (2011) argues that “System 1 thinking” (fast, emotional responses) often drives impulsive actions before rational judgment can intervene. This directly supports your finding that stress, mood, and fatigue lead to impulsive choices.

- Sterman (2000) highlights that systems produce behavior through *feedback loops*

Conclusion and Recommendations

Conclusion

The study "Rethinking Decisions: Applying Systems Thinking to Understand Patterns, Feedback, Consequences in Everyday Choices" highlights the significant role of systems thinking in enhancing personal decision-making among students. The findings indicate that most participants demonstrate awareness of patterns, feedback loops, and the consequences of their everyday choices. Students recognize that emotional states, daily habits, and social influences are interconnected, creating both reinforcing and balancing feedback loops that shape behavior over time.

Specifically, the study found that:

1. Decisions are interconnected: Choices in one area of life, such as time management, finances, or emotional responses, have cascading effects on other aspects of personal well-being.
2. Feedback loops influence behavior: Positive habits reinforce growth and productivity, while negative behaviors trigger corrective actions, demonstrating dynamic self-regulation.
3. Leverage points are identifiable: Students can influence outcomes by monitoring routines, planning strategically, and seeking supportive environments, reflecting the practical utility of systems thinking in everyday life.

4. Cause-and-effect awareness improves decision quality: Participants showed that recognizing the consequences of small, repeated actions contributes to long-term outcomes, confirming theories from Meadows (2008), Senge (2010), and Sterman (2014) regarding system structures and feedback dynamics.

Overall, the results suggest that applying systems thinking allows individuals to make more informed, reflective, and adaptive decisions. By understanding the interdependencies between actions, habits, and consequences, students can anticipate outcomes, minimize unintended negative effects, and cultivate sustainable behaviors.

Recommendations

Based on the findings, the study proposes the following recommendations to enhance the application of systems thinking in everyday decision-making:

1. Integrate Systems Thinking in Educational Programs: Schools and universities should include systems thinking workshops or modules to teach students how to recognize patterns, feedback loops, and long-term consequences in personal and academic decisions.
2. Encourage Reflective Practices: Students should be guided to reflect regularly on their choices, monitor routines, and analyze outcomes to strengthen awareness of cause-and-effect relationships. Journaling or digital tracking tools can be helpful in this process.

3. Promote Planning and Habit Monitoring: Establishing daily routines, goal-setting strategies, and reminders can serve as practical leverage points to reinforce positive behaviors and correct undesirable ones.
4. Leverage Social and Environmental Support: Encouraging supportive peer networks, mentorship, and structured group activities can enhance decision-making by providing constructive feedback loops that reinforce good habits.
5. Develop Tools for Visualizing Decisions: Educational technology platforms and apps can be used to help students visualize decision patterns, track feedback, and anticipate consequences, translating systems thinking principles into practical tools for daily life.
6. Future Research: Further studies should explore systems thinking in broader and more diverse populations, including younger students, professionals, and community members, and examine the long-term impact of systems thinking training on personal and organizational outcomes.

By adopting these recommendations, individuals and institutions can foster more holistic, adaptive, and sustainable decision-making practices, ultimately improving personal well-being, productivity, and social responsibility.

References

- Cheng, L. (2021). Identifying root causes in everyday problem-solving: A systems-thinking approach. *Journal of Behavioral Systems.
- Liu, M., & Park, S. (2021). Systems thinking and personal decision-making: Understanding interconnected behaviors. *International Journal of Cognitive Development.
- Martinez, R. (2023). Long-term thinking and everyday choices: The role of mental models in decision outcomes. *Journal of Applied Decision Science.
- Patel, S. (2021). Barriers to systems thinking in learning environments: A review of challenges and opportunities. *Educational Systems Review.
- Rahman, H., & Velasco, J. (2022). Feedback loops, reflection, and decision adjustment among young adults. *Asian Journal of Behavioral Studies.
- Santos, E., Dela Cruz, P., & Navarro, J. (2022). Understanding feedback loops in daily habits: A systems perspective. *Philippine Journal of Social Systems.
- Torres, A. (2022). Systems thinking in leadership and organizational decision-making. *Journal of Leadership and Innovation.
- Wu, Q., & Hernandez, L. (2023). Digital tools for systems thinking: Improving pattern recognition through technology. *Journal of Technology-Enhanced Learning.

APPENDICES

SYSTEMS THINKING IN EVERYDAY LIFE QUESTIONNAIRE

Researcher's Information

- **Research Title:** Rethinking Decisions: Applying Systems Thinking to Understand Patterns, Feedback, and Consequences in Everyday Choices
- **Researchers' Name:** Mhyca Joy N. Obra
Shina Marie Apolinar
- **Course & Section:** BS in Information Systems - 4A
- **Instructor:** Shalimar Navalta
- **Institution:** College of Information Systems
- **Date:** _____
- **Topic:** Decision-Making and Problem-Solving

Respondent Profile

(Please fill out the following information.)

1. **Age:** _____
2. **Gender:** ☐ Male ☐ Female ☐ Prefer not to say
3. **Occupation/Role:** _____
4. **Educational Level:** ☐ High School ☐ College ☐ Graduate ☐ Others: _____
5. **Have you heard of "Systems Thinking" before?** ☐ Yes ☐ No

Purpose of the Study

This questionnaire is part of a student research project on **Systems Thinking in Everyday Life**. The goal of the study is to explore how people understand connections, patterns, and relationships among different factors in everyday systems — such as health, education, work, or community.

Your participation will help the researcher analyze how individuals apply systems thinking principles in real-world contexts.

Instructions for Answering

1. **Read each statement carefully.**
2. **Indicate your level of agreement** by encircling or checking the number that best describes your opinion. Please answer all items as honestly as possible.
3. **Use the following scale:**

Scale	Description
1	Strongly Disagree
2	Disagree
3	Neutral / Undecided
4	Agree

Scale Description

5 Strongly Agree

4. There are **no right or wrong answers** — only your honest perspective matters.
5. Your responses will be **kept confidential** and used **solely for academic purposes**.
6. If anything is unclear, feel free to ask the researcher before answering.

Statement	1	2	3	4	5
1. I consider how different factors influence the problem before deciding.					
2. I analyze possible consequences of my decisions on others.					
3. I look for patterns in how similar problems have occurred before.					
4. I think about both short-term and long-term effects of my actions.					
5. I identify root causes instead of just treating symptoms.					
6. I recognize feedback loops that can make problems recur.					
7. I weigh how one decision may affect other parts of my life.					
8. I reflect on how emotions may bias my decision-making.					
9. I involve others to gain different perspectives before acting.					
10. I adjust decisions when new information becomes available.					
11. I think about unintended consequences of possible solutions.					
12. I map out all stakeholders involved in a decision.					
13. I use previous experiences to anticipate future outcomes.					
14. I regularly review whether my decisions achieve the desired results.					
15. I approach complex problems by analyzing the entire system.					

Thank You!

Your participation is highly appreciated.

By completing this questionnaire, you are contributing to a better understanding of how systems thinking helps us make connections and improve everyday life.